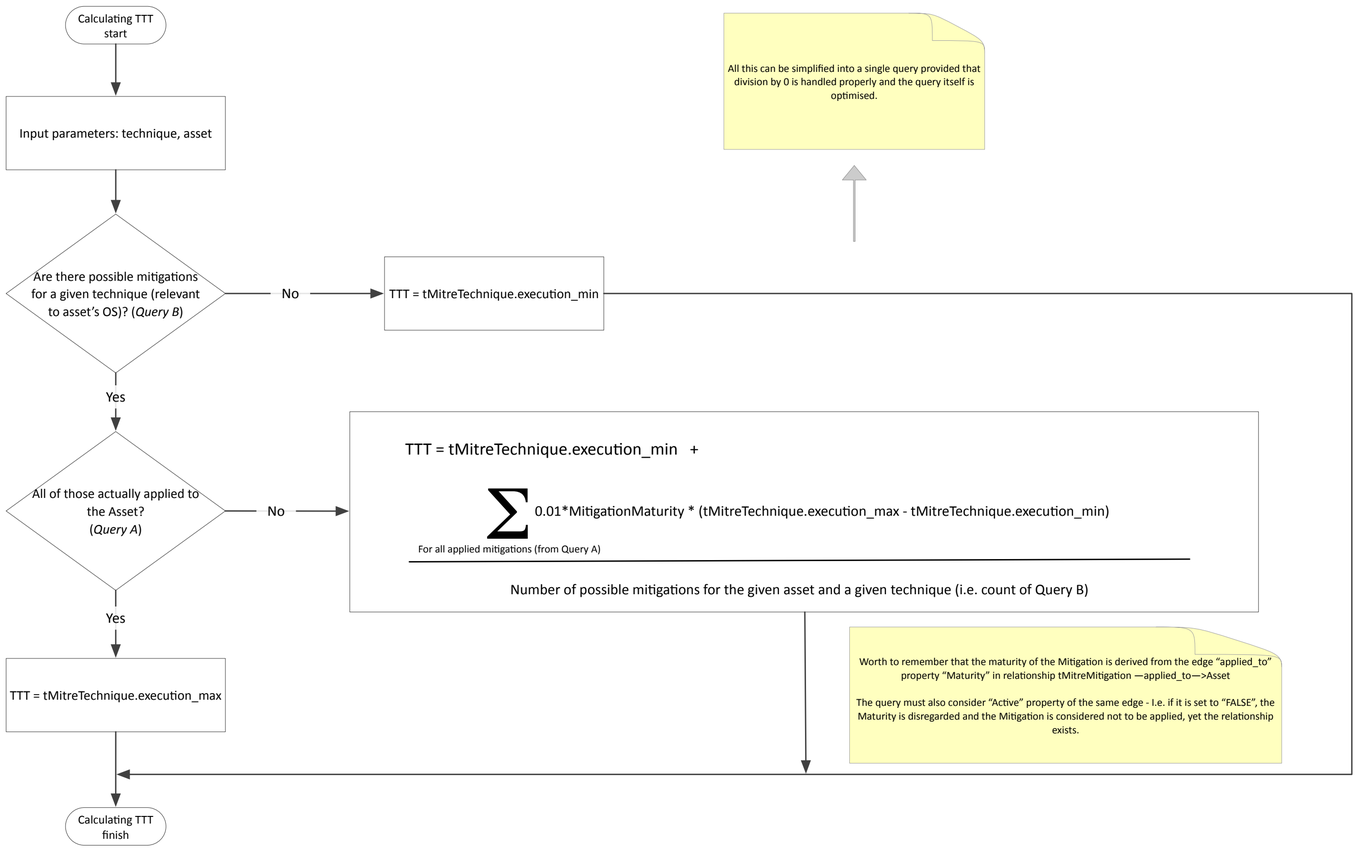
Exhibit A: TTT Calculation algorithm

TTT - Time to execuTe a Technique - time interval required to execute a technique/subtechniqe on a particular node, considering mitigations applied
(*Remember, from MITRE standpoint technique and subtechnique are similar objects except having different hierarchical relationships).*



Query B

The existing mitigations for a given technique/subtechnique (relevant to asset's OS)

```
MATCH (a:Asset)-[:runs_on]->(os:OS_Type)-[:represents]->(p:MitrePlatform)<-
[:can_be_executed_on]-(t:tMitreTechnique)<-[:mitigates]-(m:tMitreMitigation)-[:applied_to]-
>(a)
WHERE id(a) == "A00014" AND id(t) == "T1071.001"
RETURN m.tMitreMitigation.Mitigation_ID AS mitigation_id,
       m.tMitreMitigation.Mitigation_Name AS mitigation_name,
       t.tMitreTechnique.Technique_ID AS technique_id,
       t.tMitreTechnique.Technique_Name AS technique_name,
       p.MitrePlatform.platform_name AS platform
ORDER BY mitigation_id;
```

The output gives (for a given asset and a technique)

[ESP01]> MATCH (a:Asset)-[:runs_on]->(os:OS_Type)-[:represents]->(p:MitrePlatform)<-[:can_be_executed_on]-(t:tMitreTechniqu e)<-[:mitigates]-(m:tMitreMitigation)-[:applied_to]->(a) WHERE id(a) == "A00014" AND id(t) == "T1071.001" RETURN m.tMitreMitig ation.Mitigation_ID AS mitigation_id, m.tMitreMitigation.Mitigation_Name AS mitigation_name, t.tMitreTechnique.Technique_ID...

Table

mitigation_id	mitigation_name	technique_id	technique_name	platform
M1037	Filter Network Traffic	T1071.001	Application Layer Protocol: Web Protocols	Windows

Total 1 1

Query C

The possible mitigations for a given technique (relevant to asset's OS)

```
MATCH (a:Asset)-[:runs_on]->(os:OS_Type)-[:represents]->(p:MitrePlatform)<-
[:can_be_executed_on]-(t:tMitreTechnique)<-[:mitigates]-(m:tMitreMitigation)
WHERE id(a) == "A00014" AND id(t) == "T1071.001"
RETURN m.tMitreMitigation.Mitigation_ID AS mitigation_id,
       m.tMitreMitigation.Mitigation_Name AS mitigation_name,
       t.tMitreTechnique.Technique_ID AS technique_id,
       t.tMitreTechnique.Technique_Name AS technique_name,
       p.MitrePlatform.platform_name AS platform
ORDER BY mitigation_id;
```

The output gives (for a given asset and a technique)

[ESP01]> MATCH (a:Asset)-[:runs_on]->(os:OS_Type)-[:represents]->(p:MitrePlatform)<-[:can_be_executed_on]-(t:tMitreTechniqu e)<-[:mitigates]-(m:tMitreMitigation) WHERE id(a) == "A00014" AND id(t) == "T1071.001" RETURN m.tMitreMitigation.Mitigation_I D AS mitigation_id, m.tMitreMitigation.Mitigation_Name AS mitigation_name, t.tMitreTechnique.Technique_ID AS technique_id, ...

Table

mitigation_id	mitigation_name	technique_id	technique_name	platform
M1031	Network Intrusion Prevention	T1071.001	Application Layer Protocol: Web Protocols	Windows
M1037	Filter Network Traffic	T1071.001	Application Layer Protocol: Web Protocols	Windows

Total 2 1

Query D

All logic summed up in a single query

```
MATCH (a:Asset)-[:runs_on]->(os:OS_Type)-[:represents]->(p:MitrePlatform)<-[:can_be_executed_on]-(t:tMitreTechnique)
WHERE id(a) == "A00014" AND id(t) == "T1071.001"
WITH a, t,
     t.tMitreTechnique.execution_min AS exec_min,
     t.tMitreTechnique.execution_max AS exec_max,
     t.tMitreTechnique.Technique_ID AS technique_id,
     t.tMitreTechnique.Technique_Name AS technique_name
OPTIONAL MATCH (t)<-[:mitigates]-(m_all:tMitreMitigation)
WITH a, t, exec_min, exec_max, technique_id, technique_name,
     count(m_all) AS possible_count
OPTIONAL MATCH (t)<-[:mitigates]-(m_applied:tMitreMitigation)-[ap:applied_to]->(a)
WITH technique_id,
     technique_name,
     exec_min,
     exec_max,
     possible_count,
     collect({maturity: ap.Maturity, active: ap.Active}) AS applied_raw
WITH technique_id,
     technique_name,
     exec_min,
     exec_max,
     possible_count,
     [x IN applied_raw WHERE x.active == true] AS active_mitigations
WITH technique_id,
     technique_name,
     exec_min,
     exec_max,
     possible_count,
     size(active_mitigations) AS applied_count,
     CASE WHEN possible_count == 0 THEN 0.0
          ELSE reduce(s = 0.0, x IN active_mitigations | s + 0.01 * x.maturity)
     END AS maturity_factor
RETURN technique_id,
       technique_name,
       applied_count,
       possible_count,
       exec_min,
       exec_max,
       CASE
         WHEN possible_count == 0 THEN exec_min
         WHEN applied_count == possible_count THEN exec_max
         ELSE exec_min + (maturity_factor * (exec_max - exec_min)) / toFloat(possible_count)
       END AS TTT;
```

Example output

[ESP01]> MATCH (a:Asset)-[:runs_on]->(os:OS_Type)-[:represents]->(p:MitrePlatform)<-[:can_be_executed_on]-(t:tMitreTechniqu e) WHERE id(a) == "A00014" AND id(t) == "T1071.001" WITH a, t, t.tMitreTechnique.execution_min AS exec_min, t.tMitreTechniqu e.execution_max AS exec_max, t.tMitreTechnique.Technique_ID AS technique_id, t.tMitreTechnique.Technique_Name AS techni...

Table

technique_id	technique_name	applied_count	possible_count	exec_min	exec_max	TTT
T1071.001	Application Layer Protocol: Web Protocols	1	2	0.1667	120	60.083349999999996

Total 1 1